GMU ASSIP Digital Laboratory Notebook

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Location: Institute of Advanced Biomedical Research, Manassas

Sector: Department of Chemistry & Biochemistry, College of Science, George Mason University

Project: Computationally-Driven Design & Engineering of Beta-D-Glucose Binding Protein

Running MD Simulation Jobs on Hopper

1. Organize the associated files into a designated folder
2. Obtain a PDB file of just the protein through ChimeraX and input into the folder
3. Obtain a MOL2 file of the ligand through ChimeraX
4. Input this ligand.mol2 file into Charmm-GUI and download the PDB and PSF format files from the site and place it into the folder
5. Open VMD and specify the folder in its terminal | "cd *Directory*"
6. Load the PDB structure of just the protein and go to Extensions > Modeling > Automatic PSF builder
7. Load input files > Guess and split chains using current selection > Create chains
8. Load the Ligand
9. Change script.tcl with the correct psf and pdb files
10. vmd -dispdev text -e script.tcl
11. Go to Extensions > Modeling > Add Solvation Box
12. Uncheck water box and check "Use Molecule Dimensions"
13. Make all values of box padding 30
14. Extensions > Modeling > Ionization > Make sure NaCl is chosen > Neutralize and set Ion concentration.. > Autoionize
15. Open the tk console type "source maxmin.tcl"
16. Input ionized.pdb for the pdb file
17. Type all for every next prompt
18. Obtain the coordinates and input into the namd file | Then increase the PME grid size to at least one greater than the coordinates
19. On the slurm script, change lines 3, 4, 5, 25, and 27 to have the name of the job and change the file name to that same job name for both the slurm and namd files
20. Open the terminal
21. scp -r C:/Users/akhil/Downloads/ASSIP/MDSimulations/MD <NETID>@hopper.orc.gmu.edu:/scratch/<NETID>
22. Input password and the following files from the specified directory should load in the terminal
23. Sign in through ssh <NETID>@hopper.orc.gmu.edu
24. chmod u+w /scratch/<NETID>/*SimulationFolderName*
25. cd /scratch/<NETID>
26. ls | Ensure the directory is present
27. cd *directory*
28. ls | All files in the directory should appear
29. Sbatch file.slurm | Submits a batch job
30. squeue -u <NETID> | View the status of the job
31. Sacct | View jobs completed and their status (completed, cancelled, failed…)

Analyzing Results

1. Open OnDemand > Files > /scratch.<NETID>/*directory*
2. Download the DCD or similar output files
3. VMD > Load ionized.psf > for ionized.psf, load the dcd file and observe
4. Graphic > Representations
5. Create a representation of just the protein by typing "protein" in the New Cartoon drawing method
6. Create a representation of the ligand by typing "not protein and not water"
7. Run the simulation and observe
8. Extensions > Analysis > RMSD trajectory > Align & RMSD > If the angstroms are below 2 or 3, the protein is relatively stable
9. Extensions > Analysis > Hydrogen Bonds > protein 1: protein | protein 2: not protein and not water
10. Calculate detailed info for all hbonds
11. Plot data with multiplot and specify the output directory in the original folder
12. Write output to files > Find hydrogen bonds
13. Open the details file to determine the residues that interact with ligand and the corresponding extent of hydrogen bonding
14. Optimize structures with low occupancy

Running LigandMPNN Jobs on Hopper

1. Log in
2. source ~/ENTER/etc/profile.d/conda.sh
    i. If the miniconda install path is unknown, use find ~ -name "conda.sh" to determine it
3. conda activate ligandmpnn_env
4. (ligandmpnn_env) [<NETID>@login001 ~]$ | This or a similar notation should appear
5. cd ~/LigandMPNN
6. salloc -p gpuq -q gpu --nodes=1 --ntasks-per-node=12 --gres=gpu:1g.10gb:1 --mem=15gb -t 0-02:00:00
7. Reinitialize if needed:

source ~/ENTER/etc/profile.d/conda.sh
conda activate ligandmpnn_env

8. python run.py \
  --seed 111 \
  --pdb_path "./inputs/*structure*.pdb" \
  --out_folder "./outputs/default"

Make sure the input PDB file is in the ./inputs/ directory

python run.py \
  --model_type "ligand_mpnn" \
  --pdb_path "./inputs/6plpmpnn16_1.pdb" \
  --temperature 0.1 \
  --out_folder "./outputs/default" \
  --batch_size 20 \
  --number_of_batches 1

9. View results with ls ./outputs/default
    i. Seqs.csv (design sequences), scores.json (scoring data), and possible .pdb files or logs will appear
10. This data can be downloaded through scp as well:

scp your_username@hopper.gmu.edu:~/LigandMPNN/outputs/default/seqs.csv.

Running ProteinMPNN Jobs on Hopper

1. Log in
2. source ~/ENTER/etc/profile.d/conda.sh
    i. If the miniconda install path is unknown, use find ~ -name "conda.sh" to determine it
3. conda activate proteinmpnn
4. (proteinmpnn) [<NETID>@login001 ~]$ | This or a similar notation should appear
5. Cd ~/ProteinMPNN
6. salloc -p gpuq -q gpu --nodes=1 --ntasks-per-node=12 --gres=gpu:1g.10gb:1 --mem=15gb -t 0-02:00:00
7. Reinitialize if needed:

source ~/ENTER/etc/profile.d/conda.sh

conda activate proteinmpnn

8. Open OnDemand and create a ".chain_id" file with the following content:

{ "name": "*pdbfilename*", "chain_id": "A" }

9. If you need fixed residues create a "name.json1" file with the following content:

{"*pdbfilename*": {"*chainname*": [15, 17, 25, 43, 47, 53, 55, 71, 73, 76, 79, 81, 97, 99, 105]}}

10.

```
python protein_mpnn_run.py \
  --pdb_path inputs/7mpnn1_13/7mpnn1_13.pdb\
  --chain_id_jsonl inputs/7mpnn1_13/7mpnn1_13.chain_id \
  --fixed_positions_jsonl inputs/7mpnn1_13/7mpnn1_13fixed_positions.jsonl \
  --out_folder outputs/ \
  --num_seq_per_target 20 \
  --sampling_temp 0.1
```

Potential Protein Candidates

Model	FASTA	ipTM	pTM
pmpnn_12	TAAEIKALIPGTWNRTTTDENGNVATGTVTFTPVDENTW AVKQTGTRADGTPIDATGTATLQADNTLEFTLTDAKRGG VTVTGTITYVDDDLFTLTETIGDGTKITGTLTKAA	0.7784	0.9354
2pmpnn_12	TAAEIKALIPGTWNRTTVDENGNVATGTTTFTPVDDNTWAV TQTGTRADGTPIDATGTATLQDDNTLKFTLTDAKRGGVTVT GTITYVDDDHFTLTETIADGTKITGTLTKNA	0.7965	0.9346
2pmpnn_14	SKATIKKLIPGTWKRTTTDENGNVATGTTTFTPVNDDTW AVTQTGTRADGSAIDATGTATLQDDNTLKFTLTDAKRGG VTVTGTITYVDDNTFTLTETIADGTKITGTLTADK	0.7943	0.9375
2lpmpnn12_12_1	SAAEIKALIPGTWNKTSTDEAGNVATGTVTFTPIDDNTW AVTETGTRADGTPINATGTATLQDDNTLAFTLTDANDGG MKVTGTITYVDSTHFTFTETKEDGTKVTGTLTKQP	0.8110	0.9397
2lpmpnn12_12_5	SAAQIKALIPGTWNKTSTDSAGNVATGTTTFTPIDANTW AVTETGTRADGNPINATGTATLQDDNTLAFTLTDAADGG VTVTGTITYVDDNTFTFTETKADGTKITGTLTKVP	0.8043	0.9411
2lpmpnn12_12_7	SAAEIKALIPGTWNKTSTDEAGNVATGTVTFTPIDDTTW AVTETGTRADGTAINATGTATLQADNTLAFTLKDANDGG ITVTGTITYVDDNHFTFTETKEDGTKITGTLTRVP	0.8144	0.9370
2lpmpnn12_12_6	SAAEIRALIPGTWNKTSTDTAGNVATGTVTFTPVDATTW AVTETGTRADGNPINATGTATLQADNTLAFTLTDAADGG VKVTGTITYVDANTFTFTETKEDGTKITGTLTRQP	0.8121	0.9375
2lpmpnn12_12_9	SAAEIKALIPGTWNKTSTDSAGNVATGTVTFTPIDDNTW AVRETGTRADGNAIDATGTATLQADNTLAFTLTDATDGG IKVTGTITYVDDNTFTFTETKEDGTKITGTLTKVA	0.8003	0.9397
2lpmpnn12_12_8	SAAQIKALIPGTWNKTSTDTAGNVATGTVTFTPIDDNTW AVTETGTRADGTPINATGTATLQADNTLAFTLTDAADGG ITVTGTITYVDDNTFTFTETKADGTKVTGTLTKLP	0.8084	0.9397
2lpmpnn12_12_11	SAEEIKALIPGTWNKTSTDTAGNVATGTVTFTPIDDNTW AVTETGTRADGTAINATGTATLQDDNTLAFTLTDANDGG VTVTGTITYVDDNTFTFTETKADGTKVTGTLTKVA	0.8026	0.9309
2lpmpnn12_12_13	SAAQIKALIPGTWNKTTTDTAGNVATGTVTFTPIDANTW AVRETGTRADGTAIDATGTATLQDDNTLAFTLTDAAAGG QTVTGTITYVDDNTFTFTQTKADGTAVTGTLTKTP	0.8041	0.9362
2lpmpnn12_12_14	SAAEIKALIPGTWNKTSVDSAGNVSTGTVTFTPIDDNTW AVKETGTRADGTAIDATGTATLQADNTLAFTLTDANDGG IKVTGTITYVDANTFTFTETKEDGTKVTGTLTKVA	0.8095	0.9391

2lpmpnn12_12_16	SAAEIKALIPGTWNKTDTDTAGNVSTGTVTFTPIDDTTW AVTETGTRADGSAINATGTATLQADNTLAFTLTDAADGG QTVTGTITYVDDNTFTFTQTKADGTKVTGTLTRQP	0.8094	0.9406
2lpmpnn12_12_15	SAAQIKALIPGTWNKTSVDTAGNVATGTTTFTPIDDNTW AVTETGTRADGNAINATGTATLQADNTLAFTLTDAADGG VTVTGTITYVDDNTFTFTETKADGTQITGTLTKVP	0.8154	0.9438
2lpmpnn12_12_18	SAAQIKALIPGTWNKTSTDTAGNVSTGTTTFTPIDDNTW AVTETGTRADGSAINATGTATLQADNTLAFTLKDANAGG ITVTGTITYVDDTTFTFTETLADGTKVTGTLTRTP	0.8218	0.9379
2pmpnn12_12_2	SLAAIRALVPGTWARTTVDTAGNVATGTTTFTPVDDNTW AVTQTGTRADGTPIDATGTATLQADGTLAFTLTDAKRGG VTVTGTITYVDDTTFALTEVIADGTKITGTLTRLP	0.7956	0.9408
2pmpnn12_14_9	SKEKIKKLIPGTWKRETTDENGNVATGTTTFTPVDENTW AVHQTGTRADGSPIDATGTATLQEDNTLKFELTDAKRGG VKVTGTITYVNDNLFEIEETIEDGTKITGTLTKDE	0.7957	0.9371
2pmpnn12_14_12	SKETIKKLIPGTWNRTTTDENGNVATGTTTFTPIDENTW AVHQTGTRADGTAIDATGTATLQDDNTLKFTLTDAKRGG VTVTGTITYVDDNTFTLTETIADGTKITGTLTKNK	0.7957	0.9394
2pmpnn12_14_14	SKEKIKKLIPGTWKRVTTDEKGNVATGTTTFTPIDENTW AVHQTGTRADGTPIDATGTAELQDDNTLKFTLKDAKRGG VTVTGTITYVNDNLFKLEETIEDGTKITGTLTKDE	0.8030	0.9352
3lpmpnn18_12	SAAQIKALLPGTWNKTSTDTAGNVSTGTVTFTPIDDNTW AVTETGTRADGSAINATGTATLQADNTLAFTLTDANAGG ATVTGTITYVDDTTFTFTQTRADGTTVTGTLTKQA	0.8151	0.9415
3plpmpnn18_1	SKAEIKALIPGTWNKTSTNEKGDVSTGTVTFTPIDDDTW AVTETGTRADGSAIAATGTATLQADNTLAFTLTDANAGG ITVTGTITYVDDDTFTFTETLADGTKVTGTLTRTA	0.8190	0.9354
3plpmpnn18_10	SAEEIKALIPGTWAKVSVDTAGNKSTGTVTFTPIDDNTW AVKETGTRADGSAIDATGTATLQADNTLAYTLTDAKAGG ITVTGTITPVDDNTFTFTEVLADGTKVTGTLTKLP	0.8247	0.9369
3plpmpnn18_9	TAAEIRALIPGTWFKTSTDEKGNVSTGTVTFTPIDENTW AVTETGTRADGSAIDATGTATLQADGTLAFTLTDAKAGG ITVTGTITYVDDNTFTFTETLADGTKVTGTLTRAP	0.8152	0.9351
3plpmpnn18_14	SAEEIKALIPGTWYKESTNEKGEKSTGTVTFTPIDDNTW AVKETGTRADGSEIDATGTATLQEDNTLAFTLTDAKAGG ITVTGTITYVNDNLFTFTETLADGTKVTGTLTKAP	0.8213	0.9367
4plpmpnn10_1	SAATIKALIPGTWAKVSTDEKGNKSTGTVTFTPVDDNTW AVKETGTRADGTPIDATGTATLQDDGTLAFTLTDAKAGG QKVTGTITPLDATTWSFTETLADGTKVTGTLTRLP	0.8444	0.9380

4plpmpnn10_2	**SAAEIKALIPGTWAKVSTDTAGNVSTGTVTFTPVDANTWAVTETGTRADGSPIAATGTATLQADGTLAYTLTDAAAGGQTVTGTITPVDATTWTFVETLADGTVVTGTLTRLP**	**0.8567**	**0.9414**
4plpmpnn10_3	SAAEVKALIPGTWAKVSVNEKGDVSTGTVTFTPLDENTWAVHETGTRADGTPIDATGTATLQADLTLAFTLTDAKAGGKTVTGTITPVDATTWTFTETLADGTKVTGTLTRLP	0.8477	0.9418
4plpmpnn10_4	SAAAVKALIPGTWAKTSVDEAGNVSTGTVTFTPLDANTWAVKETGTRADGSPIDATGTATLQADLTLAFTLTDAKAGGQTVTGTITPVDATTWTFTETLADGTKVTGTLTRLP	0.8380	0.9406
4plpmpnn10_6	SAAAVRALIPGTWAKTSVDTAGNVSTGTVTFTPLDANTWAVTETGTRADGSPIDATGTATLQADGTFAFTLTDAKAGGQTVTGTITPLDADTFAFTETLADGTVVTGTLTRLP	0.8299	0.9211
4plpmpnn10_8	SAAAVRAAIPGTWAKVSTDEKGNKSTGTVTFTPLDADTWAVHETGTRADGSPIDATGTATLQPDLTLAYTLTDAKAGGKTVTGTITPVDATHWTFTETLADGTKVTGTLTRLP	0.8474	0.9414
4plpmpnn10_5	SAAEVKALIPGTWAKVSTNEKGDVSTGTVTFTPIDDDTWAVHETGTRADGSAIDATGTATLQEDGTLAWTLTDAKAGGKTVTGTITPVDDTTWTFVETLADGTKVTGTLVRLP	0.8405	0.9371
4plpmpnn10_7	SKADILAKIPGTWAKTSTDTDGNVSTGTVTFTPVDANTWAVTETGTRADGSPIDATGTATLQADGTFAYTLTDAKAGGKTVTGTITPVDANTFTFTETLADGTVVTGTLTRLP	0.8428	0.9122
4plpmpnn10_9	**SKETIKALIPGTWAKTSVDTAGNVSTGTVTFTPVDENTWAVTETGTRADGSAIDATGTATLQDDGTLAFTLTDAKAGGATVTGTITPVNADTFTFTEVLADGTTVTGTLTRLP**	**0.8540**	**0.9404**
4plpmpnn10_12	TLEEIKAKIPGTWAKVSVNEKGEVSTGTVTFTPLDENTWAVHETGTRADGSPIDATGTATLQEDGTLAYTLTDAKAGGRTVTGTITPLNDTLWAFTEVLADGTKVTGLLVRLP	0.8288	0.9365
4plpmpnn10_11	TAAAIKAKIPGTWAKTSTDTNGNTSTGTVTFTPADANTWNVTETGTRADGTPIAATGTATLQADGTLAYTLTDSKAGGETVTGTITPLDDNTWSFTETLADGTTVTGTLTRLP	0.8299	0.9290
4plpmpnn10_10	SKEDVLAKIPGTWAKVSTDTAGNVSTGTVTFTPIDDNTWAVHETGTRADGSAIDATGTATLQDDNTLAFELTDAKAGGKTVTGTITPVDDNHFTFEETLADGTKVTGTLTRLP	0.8443	0.9312
4plpmpnn10_14	SLAEIKAKIPGTWAKTSVDTAGNTSTGTVTFTPIDENTWAVTETGTRADGSAIAATGTATLQADGTFAYTLTDAAAGGRTVTGTITPVDDTTFTFTETLADGTTVTGTLVRLP	0.8475	0.9299
4plpmpnn10_17	SAADVKAKIPGTWAKTSTNEKGEKSTGTVTFTPLDDNTWAVTETGTRADGSAIDATGTATLQEDNTLAFTLTDAKAGGRTVTGTITPVNDTTWTFTETLADGTKVTGTLTKLP	0.8321	0.9343

4plpmpnn10_16	TAEAVKAKIPGTWAKVSVDEKGNKSTGTVTFTPIDDNTWAVKETGTRADGSAIDATGTATLQADNTLAFTLTDAKAGGKVVTGTITPVDDDTWTFTEVLADGTKVTGTLKRLP	0.8426	0.9362
4plpmpnn10_19	DAATIKALIPGTWAKTSTDEAGNTSTGTVTFTPLDANTWAVTETGTRADGSAIDATGTATLQDDGTLAYTLTDAKAGGQTVTGTITPVDADTFTFTETLADGTKVTGTLTRLP	0.8412	0.9299
5plpmpnn2_1	**TKETIKALIPGTWAKTSTNSAGAVSTGTVTFTKKDENTWAVTETGTRADGSPIAATGTATLQEDGTLAYTLTDAAAGGATVTGTITPVNATTWTFTETLADGTVVTGTLTRLP**	**0.8703**	**0.9391**
5plpmpnn2_3	SAAAIKAKIPGTWAKTSTDEAGNVSTGTVTFTKVDDTTWAVTETGTRADGTPIAATGTATLQPDGTLAYTLTDAAAGGATVTGTITPLNDTTWTFTETLADGTVVTGTLTRLP	0.8639	0.9330
5plpmpnn2_5	TAERIKALIPGTWAKTSTDSAGNVSTGTVTFTPVDDNTWAVKETGTRADGSPIDATGTATLQADGTLAFTLTDAAAGGATVTGTITPVNDTTWTFTETLADGTVVTGTLTRLP	0.8608	0.9419
5plpmpnn2_8	TAAAIKANIPGTWAKVSTDEAGNVSTGTVTFTKVDANTWAVTETGTRADGSPIDATGTATLQADGTLAYTLTDAAAGGATVTGVITPVNATTWTFTETLADGTVVTGTLTKLP	0.8638	0.9393
5plpmpnn2_7	SKAEIKALIPGTWAKTSTDENGNVSTGTVTFTPLDADTWAVTETGTRADGTPIDATGTATLQADGTLAYTLTDAAAGGATVTGTITPQNATTWTFTETLADGTVVTGTLTRLP	0.8674	0.9422
5plpmpnn2_11	SAAAIRAALPGTWAKVSTNSAGDVSTGTVTFTPVDATTWAVKETGTRADGTPIDATGTATLQADGTLAYTLTDAAAGGATVTGVITPQNADVWTFTETLADGTVVTGTLTRLP	0.8677	0.9373
5plpmpnn2_13	SAAATKALIPGTWAKTSTDTAGNVSTGTVTFTPVSADTWAVTETGTRADGSPIDATGTATLQADGTLAYTLTDAAAGGATVTGTITPVDATTWTFTETLADGTVVTGTLTRLP	0.8654	0.9401
5plpmpnn2_15	**TAAEIRALIPGTWAKVSTNEAGEVSTGTVTFTPVDENTWAVTETGTRADGTPIAATGTATLQADGTLAYTLTDAAAGGAVVTGTITPQNADTWTFTETLADGTVVTGTLTRLP**	**0.8689**	**0.9416**
5plpmpnn2_18	SAAHIKALIPGTWAKTSTDSAGNTSTGTVTFTPVDDNTWAVTETGTRADGSAIDATGTATLQADGTLAYTLTDAAAGGATVTGTITPVDDTTWTFTETLADGTTVTGTLTRLP	0.8654	0.9435
5plpmpnn2_16	**SAAEIKALIPGTWAKVSTDAAGNVSTGTVTFTPVDANTWAVHETGTRADGTPIDATGTATLQPDGTLAYTLTDAAAGGATVTGTITPQNATTWTFVETLADGTTVTGTLTRLP**	**0.8710**	**0.9434**
5plpmpnn2_20	SAAHIKALIPGTWAKTSTNSAGQTSTGTVTFTPLDDNTWAVTETGTRADGTAIDATGTATLQPDGTLAYTLTDAAAGGATVTGTITPQNDTTWTFTETLADGTTVTGTLTRLP	0.8630	0.9402

6plpmpnn16_1	**SLAAIKAQIPGTWAKTSTDAAGNVSTGTVTFTPVDENTWAVHETGTRADGTPIDATGTATLQPDGTIDYTLTDAAAGGATVTGVITPQNATTWTFTETLADGTTVTGTLTRLP**	**0.8719**	**0.9468**
6plpmpnn16_2	SLAAIKAQIPGTWAKVSTDAAGNVSTGTVTFTPVDDNTWAVHETGTRADGTPIDATGTATLQPDGTLAYTLTDAAAGGATVTGTITPLNDDVWAFTETLADGTTVTGTLTRLP	0.8684	0.9422
6plpmpnn16_8	SLERIRALIPGTWAKTSTDAAGNVSTGTVTFTPVDENTWAVHETGTRADGTPIDATGTATLQPDGTLAYTLTDAAAGGATVTGTITPLDDDVWSFTETLADGTTVTGTLTRLP	0.8701	0.9434
6plpmpnn16_5	SLEEIKALIPGTWAKTSTDAAGNVSTGTVTFTPVDDNTWAVHETGTRADGTPIDATGTATLQPDGTLAYTLTDAAAGGATVTGTITPLNDDTWSFTETLADGTTVTGTLTRLP	0.8699	0.9440
6plpmpnn16_7	SLEHIKALIPGTWAKVSTDAAGNVSTGTVTFTPVDENTWAVHETGTRADGTPIDATGTATLQPDGTLAYTLTDAAAGGATVTGVITPLNDTVWTFTETLADGTTVTGTLIRLP	0.8683	0.9454
6plpmpnn16_6	TLAAIKAQIPGTWAKTSTDAAGNTSTGTVTFTPVDANTWAVTETGTRADGSPIAATGTATLQADGTLAYTLTDAAAGGATVTGTITPQNATTWTFTETLADGSTVTGTLTRLA	0.8684	0.9455
6plpmpnn16_9	SLAEIRALIPGTWAKTSTDAAGNVSTGTVTFTPLDADTWAVHETGTRADGTPIDATGTATLQPDGTLDYTLTDAAAGGATVTGTITPLDADTWSFTETLADGTTVTGTLTRLP	0.8695	0.9436
6plpmpnn16_12	SLAEIKALIPGTWAKTSTDAAGNVSTGTVTFTPVDENTWAVHETGTRADGTPIDATGTATLQPDGTLAYTLTDAAAGGATVTGTITPQNADVWTFTETLADGTTVTGTLTRLP	0.8705	0.9443
6plpmpnn16_10	TLAAIRAQIPGTWAKTSTDAAGNVSTGTVTFTPVDATTWAVHETGTRADGTPIDATGTATLQADGTLAYTLTDAAAGGATVTGTITPLNADVWSFTETLADGTTVTGTLTRLP	0.8690	0.9436
6plpmpnn16_11	SLAEIRALIPGTWAKTSTDAAGNVSTGTVTYTPLDENTWAVTETGTRADGTPIAATGTATLQPDGTLEYTLTDAAAGGATVTGTVTPQNADVWTFTEKLADGTTVTGTLTRLP	0.8695	0.9420
6plpmpnn16_14	SLAEIKAQIPGTWAKTSTDAAGNVSTGTVTFTPLDENTWAVTETGTRADGTPIAATGTATLQPDGTLAYTLTDAAAGGATVTGTITPQDANTWTFTETLADGTTVTGTLTRLP	0.8701	0.9435
6plpmpnn16_13	SLAHIKALIPGTWAKTSTDAAGNVSTGTVTFTPVDDNTWAVHETGTRADGTPIDATGTATLQPDGTLEYTLTDAAAGGAVVTGVITPLNDTVWSFEETLADGTKVTGTLTRLP	0.8693	0.9476
6plpmpnn16_16	**TLEEIKAQIPGTWAKTSTDAAGNVSTGTVTFTPVDANTWAVHETGTRADGTPIDATGTATLQADGTLTYTLTDAAAGGATVTGTITPQNATTWTFTETLADGTTVTGTLTKLP**	**0.8711**	**0.9471**

6plpmpnn16_17	SLARIRALIPGTWAKTSTDAAGNVSTGTVTFTPLDDNTW AVTETGTRADGTPIAATGTATLQPDGTLAYTLTDAAAGG ATVTGTITPLDDTTWSFTETLADGTTVTGTLTRLP	0.8682	0.9435
6plpmpnn16_19	**SLAAIRAQIPGTWAKVSTDAAGNVSTGTVTYTPVDADTW AVHETGTRADGTPIDAEGTATLQPDGTLAYTLTDAAAGG ATVTGTVTPQDATTWTFTETLADGTTVTGTLTRLP**	**0.8711**	**0.9438**
6plpmpnn16_18	**SLAEIKAQIPGTWAKISVDAAGNVSTGTVTFTPVDATTW AVHETGTRADGTPIDARGTATLQADGTLAYTLTDAAAGG ATVTGTITPQNATVWTFIETLADGTTVTGTLTRLP**	**0.8711**	**0.9462**
6plpmpnn16_20	TLAAIKAQIPGTWAKTSTDAAGNVSTGTVTFTPVDANTW AVHETGTRADGTPIDATGTATLQADGTLAYTLTDAAAGG ATVTGTITPQNATTWTFTETLADGTTVTGTLTRQA	0.8704	0.9440